

# To Smooth or Not? When Label Smoothing Meets Noisy Labels

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## 01 Real-world scenario: training with messy labels

- **Messy labels:** Labels are messy in the wild and cross entropy overfits noise while heavy smoothing underfits.
- The goal is to choose smoothing so confidence stays high on correct predictions and low on wrong ones.
- A principled schedule for the rate helps maintain **robust confidence** under imperfect labels.

## 02 Problem setting and decision rule

- **Decision rule:** Use **LS** when estimated noise  $<$  threshold; switch to **NLS** once noise crosses it.
- Binary:  $r_{\text{opt}} = r^* - 2\epsilon$ . Multi-class:  $r_{\text{opt}} = ((K-1)r^* - K\epsilon) / ((K-1) - K\epsilon)$ .
- Warm up with **CE**, then anneal  $r$  to target; cap  $|r|$  for stability.

## 03 Motivation and thesis: when smoothing should go negative

- High noise implicitly oversmooths predictions and pushes the optimal rate negative.
- A **phase transition** from positive smoothing to negative smoothing appears as noise grows.
- Negative smoothing restores **confidence on true classes** without amplifying noise.

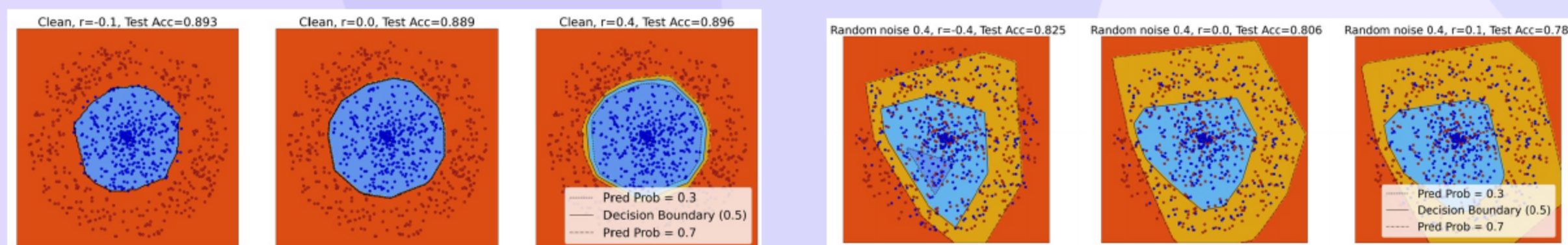
## 04 Noisy Labels and Generalized Label Smoothing (GLS)

- **Generalized label smoothing** mixes the hard label with a uniform vector using a tunable rate.
- Negative values define negative smoothing and allow extended labels that still sum to one.
- This connects to **loss correction**, **peer loss**, and **complementary labels** under clean assumptions.

## 05 Confidence Measures We Track

- Cross-entropy **confidence margin** increases with the predicted probability.
- Prediction **confidence gap** = target minus non-target probability.
- Loss-based confidence is **monotonic** in the correct-class probability.

## 06 How $r$ and Noise Shape Confidence (Intuition First)



- On clean data moderate positive smoothing works best and overly negative values overfit.
- Under symmetric noise decreasing the rate counteracts oversmoothing and improves confidence.
- At higher noise, NLS yields **higher correct** and lower wrong confidence.

## 07 Risk Decomposition: From Noisy GLS to Confidence Bias

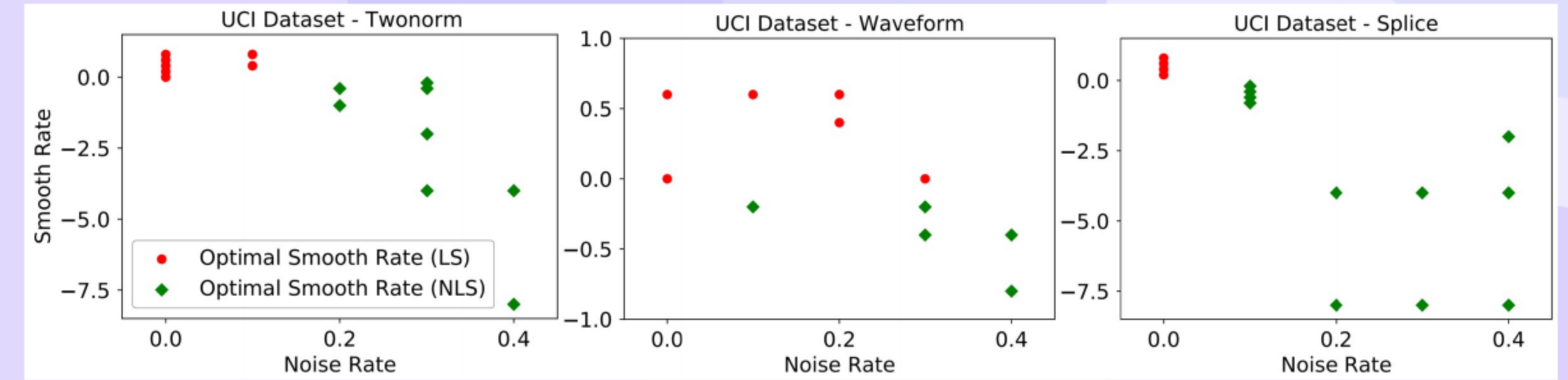
- Noisy-GLS risk = clean-GLS risk at  $r^*$  + two **confidence-bias** terms.
- The bias terms directly raise or lower the confidence margins under label noise.
- This view explains **confidence shifts** when tuning the smoothing rate under noise.



## 08 Symmetric Noise: Phase Transition in Optimal $r$

- Binary symmetric:  $r_{\text{opt}} = r^* - 2e \Rightarrow$  threshold at  $e = r^*/2$ .
- Under symmetric noise one bias term vanishes and the remaining term shifts confidence.
- The optimal rate decreases with noise and crosses zero at a noise dependent threshold.
- Below the threshold positive smoothing is preferred and above it **negative smoothing** is preferred.

## 09 Dedicated: Most surprising finding—optimal $r$ turns negative as noise grows.



- A clear phase transition shows the optimal rate turning negative as label noise increases.
- Negative smoothing undoes noise induced oversmoothing and restores useful confidence.
- This finding provides a **simple rule** to switch from LS to NLS with rising noise.
- Threshold:  $r_{\text{opt}} = r^* - 2e$  (binary), crosses zero at  $e = r^*/2$

## 10 Asymmetric noise: class-wise confidence imbalance

- Under **asymmetric noise**, a single global rate can cancel one bias term in the binary case.
- A remaining bias term shifts confidence differently across classes under imbalance.
- Tuning  $r$  trades off **class-wise confidence**.

## 11 Multi-class symmetric noise: threshold formula and intuition

- Under **multi-class symmetry** the optimal rate moves smoothly with the noise level.
- Crossing a noise dependent threshold flips from label smoothing to negative smoothing.
- Decreasing the rate counteracts implicit smoothing and **restores confidence** on true classes.



- **Warm up** with cross entropy and then anneal the rate toward a negative target.
- Use cues such as noise estimates validation stalls and loss diagnostics to trigger switching.
- Linear annealing avoids abrupt shifts and [stabilizes training](#) when noise is high.

| Method                                      | CIFAR-10, Symmetric |                  |                  | CIFAR-10, Asymmetric |                  | CIFAR-100, Symmetric |                  |
|---------------------------------------------|---------------------|------------------|------------------|----------------------|------------------|----------------------|------------------|
|                                             | $\epsilon = 0.2$    | $\epsilon = 0.4$ | $\epsilon = 0.6$ | $\epsilon = 0.2$     | $\epsilon = 0.3$ | $\epsilon = 0.4$     | $\epsilon = 0.6$ |
| Cross Entropy                               | 86.45               | 82.72            | 74.04            | 88.59                | 86.14            | 48.20                | 38.27            |
| Bootstrap (Reed et al., 2014)               | 86.06               | 81.65            | 75.26            | 87.69                | 85.51            | 47.28                | 35.81            |
| FLC (Patrini et al., 2017)                  | 84.85               | 84.98            | 73.97            | 89.42                | 88.25            | 53.04                | 41.59            |
| SCE (Wang et al., 2019)                     | 89.39               | 80.31            | 75.28            | 88.07                | 85.93            | 49.34                | 38.87            |
| APL (Ma et al., 2020)                       | 88.42               | 81.27            | 76.62            | 88.75                | 87.41            | 51.63                | 42.31            |
| Peer Loss (Liu & Guo, 2020)                 | 90.21               | 86.40            | 79.64            | 91.38                | 89.65            | 62.16                | 53.72            |
| ELR (Liu et al., 2020)                      | 92.57               | 91.32            | 88.86            | 93.48                | 92.21            | 68.03                | 60.49            |
| AUM (Pleiss et al., 2020)                   | 91.52               | 87.85            | 81.71            | 92.17                | 90.63            | 59.29                | 44.05            |
| Label Smoothing (LS) (Lukasik et al., 2020) | 90.24               | 83.78            | 75.01            | 90.61                | 88.04            | 55.17                | 41.63            |
| Negative/Not Label Smoothing (NLS)          | 89.05               | 84.85            | 77.82            | 90.02                | 88.42            | 58.47                | 46.58            |

- Label smoothing and negative smoothing are lightweight and effective under rising noise.
- State of the art robust methods are stronger in extreme noise but add complexity.
- Negative smoothing complements advanced methods and narrows gaps in [high noise](#).

- On UCI datasets the predicted optimal rate matches the **clean risk minimizer** under symmetric noise.
- As noise increases negative smoothing becomes increasingly competitive and often best.
- The observed transition aligns with the theory and yields [consistent gains](#).

| Smooth Rate        | CIFAR-10 Symmetric |                  |                  |                  |
|--------------------|--------------------|------------------|------------------|------------------|
|                    | $\epsilon = 0.0$   | $\epsilon = 0.2$ | $\epsilon = 0.4$ | $\epsilon = 0.6$ |
| $r = 0.8$          | 92.91±0.06         | 88.88±1.61       | 81.48±2.91       | 73.16±0.16       |
| $r = 0.6$          | 92.33±0.09         | 87.50±1.31       | 82.11±0.86       | 73.59±0.15       |
| $r = 0.4$          | 93.05±0.04         | 87.13±0.07       | 81.50±1.42       | 74.21±0.19       |
| $r = 0.0$          | 91.44±0.16         | 85.08±0.86       | 80.42±2.29       | 75.34±0.13       |
| $r = -0.4$         | 93.55±0.06         | 87.55±0.08       | 81.58±0.19       | 75.95±0.13       |
| $r = -0.8$         | 92.74±0.05         | 88.46±0.11       | 81.56±0.15       | 76.15±0.14       |
| $r = -1.0$         | 92.58±0.08         | 88.58±0.08       | 81.95±0.10       | 76.20±0.10       |
| $r = -2.0$         | 93.30±0.03         | 88.78±0.09       | 83.64±0.15       | 76.11±0.07       |
| $r = -4.0$         | 93.13±0.04         | 88.90±0.07       | 84.34±0.13       | 77.22±0.09       |
| $r = -6.0$         | 93.14±0.08         | 88.94±0.11       | 84.52±0.13       | 77.42±0.16       |
| $p\text{-value} =$ | 0.0004             | 0.008            | 0.011            | $< 1e - 14$      |

- On CIFAR negative smoothing leads as noise increases with strong improvements at high noise.
- Under both symmetric and asymmetric flips negative smoothing remains competitive.
- The accuracy gains are [statistically significant](#) in many high noise settings.

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Evidence on CIFAR: confidence behavior on test

- Under noise negative smoothing raises confidence for correct predictions and lowers it for wrong ones.
- Confidence distributions** reflect the intended separation between correct and incorrect cases.
- This separation produces **clearer margins** and aligns with the bias view.

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Connections: loss correction, peer, and complementary labels as NLS (when/limits).

- Equivalences:** Loss correction reduces to negative smoothing under symmetric noise with matching rates.
- Peer loss equals negative smoothing when class proportions are balanced in the training data.
- Complementary label training corresponds to an extreme negative smoothing limit.

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Practical significance on real-world noisy datasets

Results of baselines are obtained through the public leader board.

| Method     | Clothing 1M  | CIFAR-10N Aggre | CIFAR-10N Rand1 | CIFAR-10N Worse | CIFAR-100N Fine |
|------------|--------------|-----------------|-----------------|-----------------|-----------------|
| CE         | 68.94        | 87.77           | 85.02           | 77.69           | 55.50           |
| BLC        | 69.13        | 88.13           | 87.14           | 77.61           | 57.14           |
| FLC        | 69.84        | 88.24           | 86.88           | 79.79           | 57.01           |
| PL         | 72.60        | 90.75           | 89.06           | 82.53           | 57.59           |
| F-div      | 73.09        | 91.64           | 89.70           | 82.53           | 57.10           |
| LS (best)  | 73.44        | 91.57           | 89.80           | 82.76           | 55.84           |
| NLS (best) | <b>74.24</b> | <b>91.97</b>    | <b>90.29</b>    | <b>82.99</b>    | <b>58.59</b>    |

- On real world noise benchmarks negative smoothing is strong among simple baselines.
- On Clothing1M and CIFAR noisy sets negative smoothing ranks highly without extra complexity.
- The method is **simple to adopt** and pairs well with advanced robust training.

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Limitations and side-effects: overconfidence risks

- Overly negative smoothing can overfit clean data and harm calibration.
- Guardrails include warm up bounding the magnitude and close validation monitoring.
- When noise is low revert to label smoothing or vanilla loss to avoid **sharp overconfidence**.



- Use label smoothing at low noise and switch to negative smoothing as noise increases.
- Negative smoothing is simple effective and closely connected to robust loss designs.
- **Future work** extends negative labels beyond classification for [broader impact](#).

- The authors study how smoothing behaves under noisy labels and how to set the rate.
- The problem is to balance confidence on correct patterns against mislabeled data.
- The aim is [better generalization](#) with a simple choice of **smoothing rate**.

# Thank You

Questions & Discussion